

ANKIT DASH

Bhubaneswar, India | 005ankitdash@gmail.com | +91 8480212936 | linkedin.com/in/ankitdash-edu |
github.com/ankit848-ai

SUMMARY

AI/ML Developer with experience in Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Agentic AI, and multi-modal deep learning. Built systems achieving 87% cross-dataset accuracy in sleep stage detection and 99.90% in malware classification using PyTorch, XGBoost, LangChain, and HuggingFace Transformers. Oracle OCI AI Foundations certified.

TECHNICAL SKILLS

Languages: Python, SQL

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Random Forest, CNN, TCN, ResNet, Vision Transformer (ViT), U-Net, Transfer Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation

LLM & Generative AI: Large Language Models (LLM), LangChain, Retrieval-Augmented Generation (RAG), HuggingFace Transformers, Mistral-7B, Prompt Engineering, Agentic AI, Multi-Agent Systems

MLOps & Deployment: FastAPI, Docker, MLflow, REST API

Data & Analysis: NumPy, Pandas, Time-Series Analysis, EEG Signal Processing (MNE)

Tools & Platforms: Git, GitHub, Linux, Jupyter Notebook, VS Code

EXPERIENCE

Machine Learning Intern

June 2025 – July 2025

EdiGlobe

Remote

- Designed end-to-end ML pipelines covering data preprocessing, feature engineering, and model training on tabular datasets; documented experiments and results for team reproducibility.
- Built and evaluated time-series forecasting models using RMSE and ROC-AUC; iterated on model architecture based on evaluation metrics to improve generalization.

PROJECTS

Sleep Quality & Stage Detection – Multi-Modal Deep Learning + GenAI | *PyTorch, MNE, Mistral-7B, TCN* GitHub

- Trained a two-level TCN (314K params, dilated convolutions, dilation 1/2/4) on 98,985 EEG/EOG epochs from 100 SHHS1 subjects – Butterworth bandpass (0.5–45 Hz), 30 s epoch segmentation, WeightedRandomSampler for class imbalance.
- Achieved **87% cross-dataset accuracy** on 66,853 SHHS2 epochs (zero-shot SHHS1 → SHHS2 transfer); per-class F1: Wake 0.91, NREM 0.90, REM 0.70; per-subject validation on 3 unseen subjects: 86.88%, 86.16%, 71.91%.
- Integrated Mistral-7B-Instruct-v0.2 (4-bit NF4 quantized via HuggingFace) to auto-generate structured clinical PSG reports from predicted Sleep Efficiency, TST, WASO, and REM% – eliminating manual specialist interpretation per subject.

Hierarchical Agentic Malware Classification System | *PyTorch, Scikit-learn, XGBoost, ResNet18*

GitHub

- Built a 3-level agentic pipeline on MallImg (25 families, 7,459 images): RF + XGBoost on 256-dim histogram/entropy features at Stage-1, escalating low-confidence samples (threshold = 0.90, ~11% escalation rate) to a fine-tuned ResNet18 CNN at Stage-2.
- Designed a custom **DecisionEngine** with entropy-based escalation and a Logistic Regression meta-model; benchmarked all 4 classifiers – RF: 99.79%, XGBoost: 99.90%, ResNet18: 97.60%, Agentic: **99.90%** test accuracy.
- Serialized full inference pipeline via joblib + torch.save with per-sample risk scoring (Low / Medium / High) and family-level alert generation across all 25 malware classes.

Copy-Move Image Forgery Detection | *TensorFlow, ResNet50, Vision Transformer, U-Net*

GitHub

- Designed a hybrid ResNet50 + Vision Transformer (ViT, 4-head Multi-Head Attention, LayerNorm) + U-Net architecture for pixel-level copy-move forgery segmentation, trained on 5,000 paired image samples over 50 epochs.
- Achieved **92% pixel-level accuracy**, Dice coefficient 0.89, and IoU 0.80; Tampered Region F1: 0.91, Background F1: 0.92 – trained with Dice Loss + Binary Cross-Entropy and EarlyStopping / ReduceLROnPlateau callbacks.
- Post-processed segmentation masks via skimage contour detection to visually outline tampered regions on original images with dotted red contours for interpretable forensic output.

CERTIFICATIONS & ACHIEVEMENTS

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

March 2026

Oracle University – Credential ID: 326624406OCI25AICFA

EDUCATION

Centurion University of Technology and Management

B.Tech – Computer Science Engineering, Data Analytics & Machine Learning (DAML)

Bhubaneswar, India

July 2024 – July 2027

Dhableswar Institute of Polytechnic

Diploma – Computer Science Engineering

Cuttack, India

June 2021 – June 2024

Biswanath Bidya Pitha

Secondary Education (Class X)

Athagarh, Cuttack, India

2021